

SUMMARY OF QUASI EXPERIMENTAL DESIGN

QUASI-EXPERIMENTAL DESIGN:

A research design that compares groups, but doesn't randomly assign people to those groups. Tries to show cause-and-effect relationships, but with some limitations.

Example:

Comparing students who use a new teaching method (Group A) to students who use the old method (Group B), to see which method works better.

I. RETROSPECTIVE (EX POST FACTO) DESIGN :

- Collecting data from past events, behaviours, or outcomes.
- Analyzing the data to identify patterns, correlations, or causes.
- Drawing conclusions about what happened in the past.

Example:

A researcher wants to investigate the relationship between a specific medication and a particular side effect. They collect data from medical records of patients who took the medication in the past and analyze it to identify any patterns or correlations.

Pros:

- Can provide valuable insights into past events
- Often less expensive and time-consuming than experimental designs
- Can inform future research or interventions

Cons:

- Limited control over data quality and accuracy
- No ability to manipulate variables or test causality
- May be subject to biases and confounding variables

II. PROSPECTIVE QUASI-EXPERIMENTAL DESIGN

- Selecting a group of participants who have already been exposed to a particular intervention, treatment, or condition (e.g., a new teaching method).
- Collecting data from this group over a period of time (prospectively).

- Comparing the outcomes or results from this group to a comparison group that did not receive the intervention.
- Drawing conclusions about the effectiveness of the intervention.

Example:

A researcher wants to evaluate the effectiveness of a new exercise program on weight loss. They recruit participants who are already enrolled in the program and collect data on their progress over several months. They compare the results to a group of individuals who did not participate in the program.

Pros:

- Can establish cause-and-effect relationships
- Can control for some confounding variables
- Can provide valuable insights into the effectiveness of interventions

Cons:

- Lack of random assignment may lead to biases (randomization)
- Comparison group may not be equivalent to the intervention group
- May be subject to confounding variables

III. **TIME SERIES DESIGN :**

- Collecting data from a single group or individual over a long period of time.
- Measuring the same variables repeatedly at multiple points in time.
- Analyzing the data to identify patterns, trends, or changes over time.

Example:

A researcher wants to study the daily temperature of a city over a year to identify seasonal patterns. They collect temperature data every day for a year and analyze it to identify changes in temperature over time.

Pros:

- Allows for analysis of patterns and trends over time.
- Can identify changes and fluctuations. (can study changes over time)
- Can be used to predict future behavior.

Cons:

- May be influenced by external factors.
- Requires consistent data collection.

- May be difficult to establish causality.

IV. EVENT-SPECIFIC ALIGNMENT DESIGNS :

- Identifying a specific event or intervention (e.g., a policy change, training program, or natural disaster).
- Collecting data before and after the event.
- Analyzing the data to determine if the event had an impact on the outcome or behavior being studied.

Example: A researcher wants to study the impact of a new law on crime rates. They collect crime data from before and after the law was enacted. They compare the data to determine if the law had an effect on crime rates.

Pros:

- Allows for causal inference
- Can establish cause-and-effect relationships
- Can inform policy or practice decisions

Cons:

- May be influenced by confounding variables
- Requires accurate data collection
- May be difficult to establish a clear cause-and-effect relationship

V. TWIN STUDIES AND ADOPTION STUDIES:

Twin studies and adoption studies are research designs used in behavioural genetics to investigate the role of genetics and environment in shaping behavior, traits, and disorders.

Twin Studies:

- Compare the similarity of identical (monozygotic) twins, who share 100% of their genes, to fraternal (dizygotic) twins, who share 50% of their genes on average.
- Aim to estimate the heritability of a trait, which is the proportion of variation in the trait that can be attributed to genetic factors.

Adoption Studies:

- Compare the similarity between adopted children and their biological parents, who share genes but not environment, to the similarity between adopted children and their adoptive parents, who share environment but not genes.
- Aim to estimate the heritability of a trait and the influence of environmental factors.

Both designs help researchers:

- Disentangle the effects of genetics and environment
- Understand the causes of individual differences
- Inform strategies for prevention, intervention, and treatment

Key aspects:

- **Twin studies:** Compare identical and fraternal twins
- **Adoption studies:** Compare biological and adoptive parents
- **Both:** Estimate heritability and environmental influence